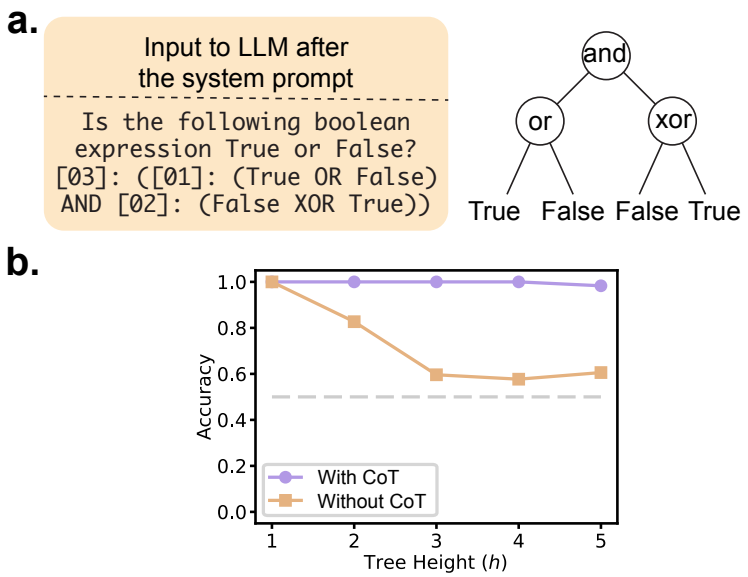

Figure 2: **a.** Example of a Boolean logic tree with height $h = 2$. *Left*: The corresponding text input provided to the model, where internal nodes are labeled [01] through [03]. *Right*: Schematic of the tree structure. **b.** Accuracy on Boolean logic trees of varying height. Performance with CoT (purple) remains near-perfect ($> 98\%$), while standard prompting (No CoT, orange) degrades significantly as tree depth increases (61% for $h = 5$). The dashed line marks the random baseline (50%).

The system prompt is designed to minimize formatting variance, allowing us to use the structural strings as *temporal anchors* for aligning representations across different tasks. We use the following condensed system prompt:

System Prompt for Structured Reasoning

You are a precise Boolean logic solver. You will be given a boolean expression where specific operations are labeled with IDs like `[01]`, `[02]`, etc.
Your task is to solve the tree step-by-step in strictly **increasing order** of these IDs. Follow the exact format shown in the example below.
EXAMPLE OUTPUT:
Problem Statement
1. **Expression**: ‘[03]: ([01]: (True or False) and [02]: (False xor True))‘
2. **Node IDs**: [01], [02], [03]
Solve
Node [01]
* Logic: ‘True or False‘
* Result: ‘True‘
Node [02]
* Logic: ‘False xor True‘
* Result: ‘True‘
Node [03]
* Logic: ‘[01] and [02]‘ → ‘True and True‘
* Result: ‘True‘
Summary
* [01]: True
* [02]: True
* [03]: True
Final Answer: True
YOUR TURN:

This structured output serves a dual purpose: it improves model accuracy (achieving 98% accuracy on depth-5 tasks with *Ministral 3 8B Reasoning*) and provides deterministic textual markers to compute the manifold capacity $\alpha_t^{(n)}$ at semantically equivalent points in the reasoning chain. While such a precise description of the task to the model arguably simplifies the reasoning the model has to do, it allows us to focus more clearly on the memory management aspect of the task.

3.3 Model Configuration

For all experiments, we utilize the *Ministral 3 8B Reasoning* model Liu et al. [2026], a state-of-the-art small language model specifically optimized for complex, multi-step reasoning tasks. The model consists of 32 transformer layers and its residual stream is of dimension 4096. Note that the model family has been distilled from Mistral Small 3.1. To ensure deterministic reasoning and stable manifold geometry across all tasks, we set the temperature to $T = 0.0$ and disable sampling. All experiments are executed on NVIDIA A100 (80GB) GPUs, utilizing the eager attention implementation for consistent activation capturing across the residual stream.

3.4 Structural Token Anchoring and the Contamination Problem

A primary challenge in analyzing neural manifolds during CoT is the issue of *textual contamination*. Because the model's response explicitly contains the answers to intermediate sub-problems, any measurement of the encoded state taken at the exact token where a solution is articulated (e.g., the token for "True" in "Node [01] Result: True") would be trivial. Such a signal would merely reflect the model's *reading* of its own generated text rather than its internal reasoning process.

To circumvent this, we restrict our manifold capacity calculations and linear probing to specific **structural anchors**. These anchors are formatting strings (e.g., `### Solve`, `* Logic:`) that precede the actual solution of a node. We focus our analysis on the **final token** of each structural string for three primary reasons:

1. **Causal Independence (Leakage Prevention):** By measuring capacity at the token right *before* the textual solution is generated, we ensure that the linear separability we observe is an emergent property of the latent representation rather than a reflection of the output token itself.
2. **Contextual Aggregation:** The last token of a structural string (e.g., the colon in `* Result:`) serves as the point of maximum contextual pressure where the model must aggregate information to produce the next logical result.
3. **Semantic Alignment:** Using fixed formatting strings provides a common time, allowing us to align trajectories across tasks that generate a varying number of tokens.

This methodology allows us to focus our attention on the internal computation.

3.5 Analysis Methods

To solve the task it is asked, the model has to successively determine the truth value of each node. At any point in the CoT generation, at any layer in the model, one can group the activations with respect to the truth value of any node in order to decode the extent to which the information about that node's value is being encoded. This decoding can be done through the lens of manifold capacity theory or with a linear probe. The tokens along the CoT generation that we analyze are the structural tokens described in the above subsection.

Manifold Capacity We measure the manifold capacity on a set $\mathcal{M} := \{\mathbf{x}_i\}_{i=1}^D$ of intermediate layer residual stream embeddings aligned at a single structural token across D number of Boolean logic problems, with classification labels $\mathcal{Y} := \{y_i\}_{i=1}^D$ specified as the ground truth answers to a single node in the Boolean logic tree.

The MCT tells us that N^*, the minimal random subspace dimensions needed for linear separation according to some labeling, can be efficiently computed by [Chou et al., 2024]

$$N^* = \left\langle \|\Pi_{\mathcal{V}}(\mathbf{t})\|_2^2 \right\rangle_{\mathbf{t}\sim\mathcal{N}(0,\mathbf{I}_N)},$$

where $\Pi_{\mathcal{V}}(\mathbf{t})$ is a Euclidean projection of the $\mathbf{t}$ vector to a cone $\mathcal{V}$ of the points $\{y_i\mathbf{x}_i\}_{i=1}^D$ where y_i is 1 if the ground truth answer of a given node is True, and -1 if False. In our work, we simply set $P = 2$ when computing α, i.e. $\alpha = 2/N^*$.

Linear Probes To complement the capacity measure, we employ linear probes trained to separate the same manifolds. We utilize two distinct linear classifiers: a Hard-Margin Support Vector Machine (SVM) and Logistic Regression. Given that the high-dimensional representations in our analysis are linearly separable, the Hard-Margin SVM serves as a direct geometric probe.

Note that the training of the hard-margin SVM yields a maximum margin hyperplane. The direction of a normal vector to this hyperplane can be used as a representative direction of the encoding of the truth value of the node in question at

that particular time. This then allows us to look at whether this direction for a given node is conserved over time, and to look at cosine similarity measures of the direction across nodes (for an accordingly chosen time, which should be node-dependent).

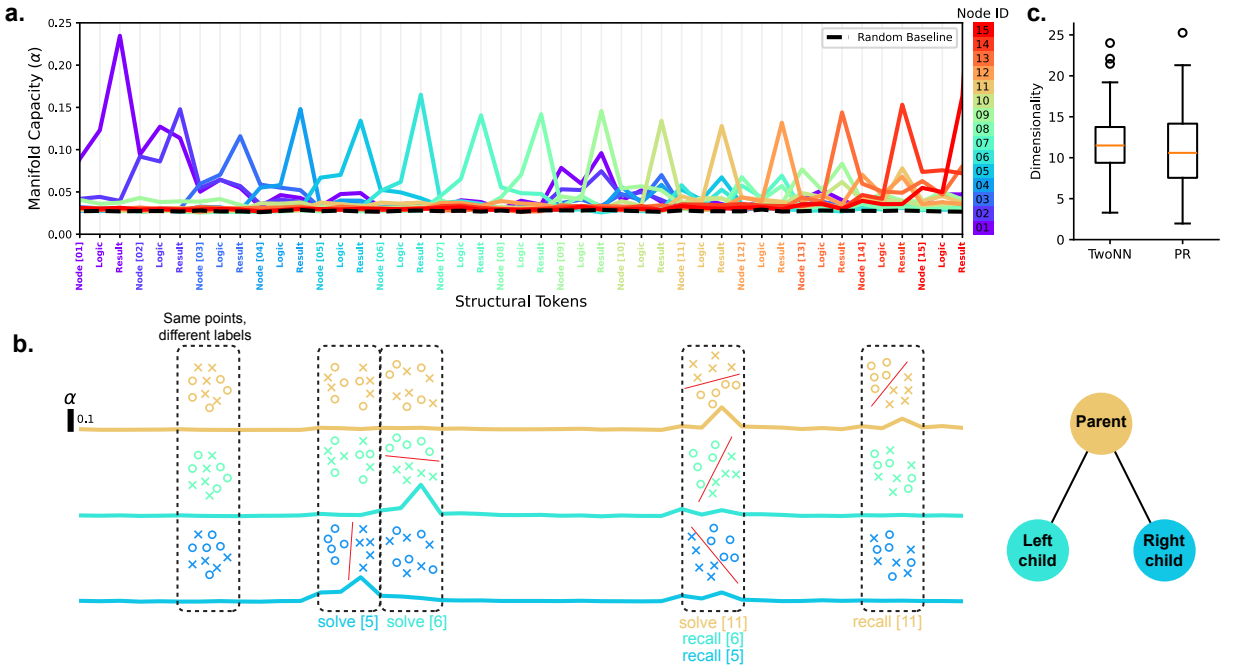

Figure 3: **Dynamic Modulation of Manifold Geometry during CoT. a.** Manifold capacity (α) tracked across the Chain-of-Thought sequence (layer 20, tree height $h = 4$, see Supplementary for height 5). Lines are colored by node ID. Capacity for a specific node peaks sharply at two distinct moments: first when the node is intrinsically computed, and second when it is processed by its parent. **b.** Detailed analysis of Node 11 (yellow) and its children, Node 5 (blue) and Node 6 (turquoise). Insets visualize the latent geometry at key timesteps. High capacity peaks correspond to linearly separable manifolds (clear decision boundaries). Notably, representations become entangled (low capacity) in the interim periods between the initial solution (solve) and the subsequent retrieval (recall). **c.** Intrinsic dimensionality estimates of the latent embeddings using Two-Nearest Neighbors (TwoNN) and Participation Ratio (PR).

4 Results

We analyze the geometry of the residual stream during reasoning on our tree boolean logic task. We first validate the model's performance, then track the evolution of Manifold Capacity and Probe Accuracy over time (tokens) and space (layers).

4.1 Task Performance

On the depth-5 Boolean Logic task, `Ministral 3 8B Reasoning` achieves 98% accuracy with full CoT generation, compared to 59% (barely above random guess) when prompted for an immediate answer (cf. Fig.2b). This confirms that CoT is functionally necessary for this architecture to solve the task, and ensures that our subsequent analysis reflects correct reasoning traces.

4.2 Dynamic Manifold Management

We track the representation geometry through the generation of the CoT through the lens of manifold capacity and linear probes. Through this we aim to exhibit a signature of the reasoning process taking place and unveil specific aspects of its dynamics.

We first look at the evolution of our metrics at layer 20 of the model, as it is in the mid to late layers that it is thought most of the reasoning actually takes place. The next subsection attends to a more specific layer $\times$ time analysis.